

CHAEWON YOON

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RESEARCH FOCUS

Efficient and reliable machine learning under constraints on precision, memory, and compute; model compression and quantization; low-precision and adaptive inference; efficient ML systems; neural operators and scientific machine learning.

EDUCATION

University of Wisconsin-Madison - Undergraduate Student (Junior), intended B.S. in Data Science *Expected May 2028*

Transferred Fall 2026. Relevant coursework: **MATH 345 Linear Algebra & Optimization** (in progress); **STAT 240 Data Science Modeling I** (in progress); COMP SCI 300 Programming II; COMP SCI 240 Introduction to Discrete Mathematics; Calculus I-II; Probability & Statistics; Applied Statistics; Independent Study in Numerical Methods; Independent Study in Neural Methods for PDEs.

University of Wisconsin Oshkosh - Undergraduate Study

Sep. 2024-May 2026

Completed foundational coursework and faculty-mentored independent studies before transferring to UW-Madison.

SELECTED PUBLICATIONS & RESEARCH MANUSCRIPTS

Peer-Reviewed Publication

- Yoon, C. "EvidenceLens: A Declarative Evidence-State Interface for Human-AI Visual Inspection." ACM UIST Adjunct 2026 (Poster). [DOI](#) | [GitHub](#)

Selected Manuscripts Under Peer Review

- Yoon, C. "Conditional Precision Bottlenecks and Quantized Defect Correction in Fractional Neural Operators." 2026. [Preprint](#) | [GitHub](#) | [Software DOI](#)
 - Yoon, C. "Transition-Preserving State Tracking via Definition-Grounded Compilation." 2026. [Preprint](#) | [GitHub](#)
 - Yoon, C. "Evidence Is Not Free: Constrained Template Evidence Acquisition for Risk-Aware Governance." 2026. [Preprint](#) | [GitHub](#) | [Software DOI](#)
- Additional manuscripts and research artifacts are available on GitHub.

FACULTY-MENTORED RESEARCH EXPERIENCE

Undergraduate Researcher (Independent Study) - Scientific ML & Neural Operators

Sep. 2025-Jul. 2026

Faculty mentor: Kamran Kazmi, Professor of Mathematics, University of Wisconsin Oshkosh

- Developed Quantized Defect Correction (QDC) for precision-sensitive failures in low-bit neural operators using Python/PyTorch, numerical PDE methods, residual analysis, controlled ablations, matrix-free LOBPCG, higher-resolution experiments, RealPDEBench data, and hardware-oriented evaluation.
- Designed and executed self-directed research cycles on fractional PDEs and Fourier neural operators, presenting research plans, code, results, and revisions for faculty critique.
- Studied numerical PDEs, error and convergence analysis, interpolation, and numerical differentiation/integration; continued the work beyond the credit-bearing independent study through the July 2026 WiSys SPARK Symposium.

Undergraduate Researcher - Stochastic Processes & Vibration Modeling

Sep. 2025-May 2026

Faculty mentor: Ahmed O. Nasif, Associate Professor of Electrical Engineering Technology, University of Wisconsin Oshkosh

- Studied stochastic processes and probabilistic inequalities; independently implemented Python simulations, equation derivations, ODE and multi-DOF vibration models, Hamiltonian formulations, matrix/eigenvalue analysis, physics-constrained neural networks, and convergence/error analysis.
- Presented "Numerical Stability and Convergence of Physics-Constrained Neural Networks in Vibration Modeling" as an oral research talk at the 2025 Pi Mu Epsilon Undergraduate Regional Mathematics Conference.

RESEARCH PRESENTATIONS

- "Placement-Aware Quantization for Fractional-PDE Neural Surrogates." WiSys SPARK Symposium, 2026. Poster presentation. [DOI](#)
- "Numerical Stability and Convergence of Physics-Constrained Neural Networks in Vibration Modeling." Pi Mu Epsilon Undergraduate Regional Mathematics Conference, 2025. Oral presentation. [DOI](#)

TECHNICAL SKILLS

Programming & ML: Python (4+ years); PyTorch; TensorFlow; NumPy; SciPy; pandas; scikit-learn; matplotlib.

ML / Research: model quantization and low-precision inference; neural operators; scientific ML; numerical PDEs; residual correction; robustness and ablation design; model evaluation.

Research Engineering: Git/GitHub; reproducible experiment pipelines; tests; experiment configuration; result ledgers; Zenodo archival releases; LaTeX; technical writing and presentation.

TEACHING, LEADERSHIP & SELECTED HONORS

- Center for Academic Resources Math Tutor, UW Oshkosh (Sep. 2025-May 2026): tutored 10-15 undergraduate students weekly across calculus, college algebra, precalculus, and linear algebra.
- AI Club: Vice President (Spring 2026) and Treasurer (Spring 2025-May 2026); organized paper discussions, workshops, and interactive AI/math activities.
- Culver Business Model Competition - 1st Place, CurbMap 360 (2025); project later featured by UW Oshkosh News and Universities of Wisconsin.
- Calculus Prize Exam Winner (2025); William Lowell Putnam Mathematical Competition competitor (2024).